Vedansh Thakkar, MS | **Artificial Intelligence Engineer** | vthakkarumn@gmail.com | +91 92036-05108 | MP, India | LinkedIn | Website

AI Engineer with **6+ years** of experience in **training** and **deploying** machine learning models for **predictive** and **generative AI** tasks using **deep** learning and **large** language models. Presented, published and **reviewed** manuscripts in international conferences and peer-reviewed journals. Skilled in transforming open-ended AI research into scalable, **production-grade** solutions for healthcare and industrial robotics clients.

PUBLICATIONS

Thakkar, V., Silverman, G. M., Kc, A., Ingraham, N. E., Jones, E. K., King, S., Melton, G. B., Zhang, R., & Tignanelli, C. J.** (2025). *A comparative analysis of large language models versus traditional information extraction methods for real-world evidence of patient symptomatology in acute and post-acute sequelae of SARS-CoV-2*. PLOS ONE, 20(5), e0323535. https://doi.org/10.1371/journal.pone.0323535

WORK EXPERIENCE

***Data Scientist* | Yash Technologies** — Dec 2025 – Present — MP, India

- Architecting and implementing tool-using **AI agents** with orchestrated multi-step workflows, incorporating retrieval-augmented generation (**RAG**), function calling, planning/execution loops, and memory/state management.
- Integrating agents with enterprise data sources and internal systems via secure, auditable APIs, **databases**, and services, enforcing least-privilege access and policy-aligned data handling.
- **Deploying** AI-enabled services and supporting **APIs** to OpenShift (Kubernetes) environments, implementing production-ready practices including health checks, resource limits, secrets management, and CI/CD pipelines.
- Applying production engineering best practices to **LLM**-based systems- latency optimization, retry logic, fallbacks, rate limiting, **guardrails**, and comprehensive monitoring- ensuring reliability and operational readiness.

***AI Engineer* | Neocortex Robotics** — Feb 2022 – Oct 2025 — Remote, USA

- Led development of multimodal **RAG** systems using CLIP, PGVector, and GPT, enabling autonomous product recognition and robotic decision-making.
- Architected and deployed **large-scale computer vision pipelines** for segmentation and detection (Detectron2, DeepLabV3, YOLO, Mask-RCNN) powering industrial pallet inspection across multiple customer sites.
- Designed **end-to-end multimodal data infrastructure**, integrating embedding pipelines, **vector search**, and LLM reasoning into production robotics workflows.
- Owned **production** ML deployments on **AWS** using, REST APIs, and CI/CD, ensuring low-latency, fault-tolerant robotic vision inference.
- Drove accuracy, latency, & system reliability improvements through dataset optimization, model tuning, & GPU **performance** engineering.
- Partnered with cross-functional robotics and automation teams to integrate CV/LLM outputs directly into **real-time robotic control systems**, reducing manual QA and improving automation throughput.

***AI Manager* | Department of Surgery, University of Minnesota** — Mar 2023 – July 2025 — Remote, USA

- Developed clinical NLP systems using **LLMs** and prompt engineering to **automate** information extraction, across various patient cohorts.
- Built end-to-end pipelines involving clinical note preprocessing, LLM-based summarization, **multi-label classification**, and severity scoring frameworks, integrating **transformer** models with domain-specific medical ontologies and **annotation** guidelines.
- Engineered a multimodal deep learning model for rib fracture detection by combining deep learning-based **medical imaging** models and structured **EHR** features using a late-fusion architecture.
- **Directed** clinicians, data engineers, and researchers to deploy reproducible pipelines using PyTorch, Transformers, **FastAPI** and AWS, contributing to **peer-reviewed** publications and **clinical decision-support** research.
- Translated research prototypes into **production-ready** ML tools integrated with hospital systems for client-side deployment.

***Machine Learning Research Engineer* | TechMah Medical** — Jan 2020 – April 2022 — Knoxville, TN, USA

- Developed deep learning models for **medical image analysis** & 3D bone reconstruction.
- Developed predictive models for orthopedic diagnostics aligned with ethical AI principles in regulated **healthcare environments**.
- Deployed explainable computer vision models for **3D** bone **reconstruction** within FDA-regulated hospital systems.
- Converted proof-of-concept models into **deployable** tools using high-dimensional medical imaging data in FDA-regulated settings.

PROJECTS

LLM-Based Multi-Stage Clinical NLP System for Symptom Detection and Severity Assessment | University of Minnesota

- Built a two-stage clinical NLP pipeline for **longitudinal** VNS and COVID patients, combining LLaMA-based **fine-tuning** for symptom presence classification and LLM **prompting** for severity assessment.
- Fine-tuned LLaMA-3.2 (**1B/3B**) and LLaMA-3.3-**70B** models to classify **13 symptoms** as present/absent/negated using parameter-efficient training on >**1,400** annotated clinical notes from five corpora (COVID, PASC, N3C, NSQIP, VNS).
- Developed a standardized severity scoring framework (mild/moderate/severe) and used structured prompting with **LLaMA-70B** to perform severity extraction from summarized clinical notes.
- Achieved high accuracy on VNS test sets (e.g., seizures **F1 = 0.98**; self-harm **F1 = 0.99**), with robust performance across gender and racial **subgroups** in **fairness** analysis.
- Conducted **internal** and **external** validation across three corpora (UMN-COVID, UMN-PASC, N3C-COVID), performing macro-averaged evaluation and demographic fairness analyses.
- **Led authored** a peer-reviewed publication in PLOS One demonstrating strengths and limitations of LLMs vs. rule-based pipelines for clinical information extraction.
- Demonstrated feasibility of automated, longitudinal **symptom + severity tracking** to support clinical decision-making in neurological/psychiatric patients undergoing VNS.

Multi-modal Deep Learning Diagnostic Model for Adult Rib Fractures | Department of Surgery, University of Minnesota

- Developed a **multimodal** deep learning pipeline for automated rib fracture **diagnosis** using chest radiographs and structured EHR data, integrating **DenseNet-121** imaging features with clinical variables through a **late-fusion** architecture.
- Implemented automated rib-region localization using a **ResNet-101** ROI extractor and standardized **DICOM** preprocessing (windowing, resizing, normalization) across >**21k** images from **35 U.S.** hospitals in **four** major US health systems.
- Trained, tuned, and validated multimodal models on **21,658 patient encounters**, achieving strong internal performance (**AUC 0.90** at MHFV; **0.80** at UFL) and robust external generalization (**AUC 0.75** at IU; **0.70** at UNC).
- Demonstrated that multimodal fusion significantly outperformed image-only baselines (e.g., **AUC 0.90 vs. 0.77**, **AUPRC 0.64 vs. 0.40**), with high negative predictive values (92–96%) supporting reliable exclusion of non-fracture cases.
- Performed model fairness and subgroup analysis showing **no significant performance differences** across gender or racial groups ($p > 0.39$), satisfying accuracy-parity and equalized-odds criteria.
- Showed the clinical utility of multimodal AI as a reliable, scalable trigger for **evidence-based rib** fracture **decision-support** pathways across diverse health systems.
- Worked closely with **clinicians** and product teams to productionize model outputs for integration into clinical workflows.

Agentic Multimodal RAG for Image-Based Product Reasoning using LangChain + PGVector | Neocortex Robotics

- Designed and implemented a Visual Retrieval-Augmented Generation (**RAG**) system using **LangChain**, enabling LLMs to **reason** over images of pet-food packaging.
- Built a custom CLIP-based image embedding pipeline and stored millions-scale embeddings in **PostgreSQL** + **PGVector** for fast cosine similarity search.
- Developed LangChain retriever class performing vector search on images & returns high-quality metadata for downstream LLM reasoning.
- Integrated the retrieval module into an **Agentic AI** pipeline, where an OpenAI / GPT model orchestrates perception, retrieval, and reasoning steps **autonomously**.
- Implemented an end-to-end agent with **tools** for image understanding, database retrieval, and **multi-step** analysis, producing accurate brand/product identification.
- Delivered a **production-ready** system enabling automated SKU classification from packaging images, **improving** accuracy and **reducing** human labeling effort.

Development of Deep Learning Models for Robotic Vision Guidance | Neocortex Robotics

- Fine-tuned various deep neural networks such as Mask RCNN, DeepLab, PointRend, YOLO, for various **instance segmentation** tasks.
- Manually annotated 6K RGB and Depth images for model training. Model detects objects for automated **robotic pick-and-place** tasks.
- Deployed models at various order fulfillment companies like PetSmart. Used **Amazon S3**, accessed via Flask API, for **model deployment**.

Medical Image Segmentation & 3D Bone Pose Regression for SSM based 3D Reconstruction | TechMah Medical

- Fine-tuned a variant of **U-Net** for knee joint **segmentation**. Utilized ~250 2D radiographs manually labeled using VIA annotation tool.
- Converted high-resolution image data into structured inputs for pose **regression models**, contributing to scalable orthopedic AI tools.
- Trained on GCP using 4 NVIDIA V100 **GPUs in parallel**. Developed a 5-layer custom CNN for 3D pose regression of knee joint.
- Utilized 316K 2D binary projections generated by 3D bone models. Labels were x, y, z rotation angles with respect to the x-ray sensor ranging from [-15,15] degrees with 3 degrees increment. **Deployed** models at the University of Tennessee hospitals systems.

SKILLS

Programming & Tools: Python, SQL, R, MATLAB, C++, Stata, Git, Docker, Linux, Windows, Tableau, Power BI

AI / ML Frameworks: PyTorch, TensorFlow, Keras, JAX, Scikit-learn, Hugging Face, LangChain, Monai

Data Engineering & Pipelines: Databricks, PySpark, ETL Pipelines, Data Lakehouse, Delta Tables, Airflow, PGVector, FAISS, Snowflake (basic), Data Preprocessing, Feature Engineering

Cloud & MLOps: AWS (S3, EC2, Lambda), GCP, Azure, CI/CD Pipelines, MLflow, Kubernetes, Flask REST APIs, API Integration, Model Deployment, Observability & Monitoring

Generative AI & Agents: LLMs (LLaMA, Mistral, GPT), RAG Systems, Agentic Workflows, SQL Agents, MCP Tools, Prompt Engineering, Fine-Tuning (LoRA, PEFT), Evaluation & Benchmarking

Computer Vision: OpenCV, CLIP, Vision Transformers, Object Detection (YOLO, Mask R-CNN, DeepLab), Segmentation, 3D Reconstruction, Anomaly Detection

Healthcare AI & Compliance: FHIR, EPIC Systems, EHR Data Integration, Explainable AI (XAI), Regulatory Compliance (GxP, FDA Standards), Data Governance & Security

EDUCATION

Master of Science | Electrical Engineering & Computer Science | Dec 2019 | Nashville, TN
Vanderbilt University (3.93/4.00)

Bachelor of Science | Electronics & Electrical Engineering | May 2017 | MP, India
Rajiv Gandhi Technological University